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Roll No.

220743

4th Sem / Civil

Subject : Water Supply and Waste Water Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Dead end method of distributing water is also known as _____
- a) grid iron system b) Ring System
c) Radial system d) Tree system
- Q.2 Coagulants are used when turbidity of water is more than _____
- a) 50 ppm b) 70 ppm
c) 60 ppm d) 80 ppm
- Q.3 Filtration and sewage forming are methods of _____
- a) Land treatment b) Water treatment
c) Self purification d) All of these

- Q.4 The water which is not chemically pure but does not contain any harmful contents to human health is called _____
- a) Wholesome water b) Potable water
c) Palatable water d) Polluted water
- Q.5 The earth water circulatory system is known as _____
- a) Water cycle b) Transpiration cycle
c) Hydrological cycle d) Both a & b
- Q.6 The excess water from rain water, which run off on the surface of ground is called _____
- a) Waste water b) Drain water
c) Storm water d) None of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 _____ sludge is a sludge which remains for a longer period in the tank before removal. (Activated/ Septic)
- Q.8 The ratio of the maximum daily consumption to the average daily demands is _____ (1.0/1.8)
- Q.9 In a water distribution network _____ type of valve works automatically.(check valve / Sluice valve)

- Q.10 The velocity of sewer is responsible for floating of suspended matters is called _____ (Self cleaning velocity / Non-scouring velocity)
- Q.11 At about _____ percent moisture, sludge becomes dry. (20% / 10%)
- Q.12 Water supply scheme is usually designed for a life of _____ (75 years / 50 years)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Describe any five various factors affecting the per capita demand of water for a city.
- Q.14 Describe the use of “Stabilization Ponds” for the wastewater treatment to ensure the safe disposal of waste materials into the environment.
- Q.15 Write a short note on “Coagulation Process” of water treatment.
- Q.16 Explain the process of “Anaerobic Digestion” used for activated sludge treatment and disposal of sewage system.
- Q.17 Explain the working process of water treatment by “Reverse Osmosis System”.
- Q.18 Enlist various physical test conducted on water and describe any one of them in brief.

- Q.19 Write any five advantages of elevated reservoirs used for water distribution system.
- Q.20 Enlist any five necessity of sanitation.
- Q.21 Explain the working principle of water meter used in domestic water supply in India.
- Q.22 Describe the function of “Skimming Tanks” used for sewage treatment.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the method of “Disinfection of Water” with the help of graphical representation including break point chlorination.
- Q.24 Explain the working of “Grid-Iron Water Distribution System” along with its advantages with the help of a diagram.
- Q.25 Explain the “Disposal by Dilution” method of disposal of sewage.